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# Gaussian Elimination

## AIM:

To write a program to find the solution of a matrix using Gaussian Elimination.

## Equipments Required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

## Algorithm

1. Convert the given system of equations into an upper triangular matrix by eliminating variables step by step.
2. Swap rows if necessary to avoid division by zero and improve numerical stability.
3. Solve for unknowns starting from the last equation (bottom row) and substitute upwards.
4. Collect the computed values into a solution vector representing the final answer.

## Program:

```
/*
'''Program to solve a matrix using Gaussian elimination without partial pivoting.
Developed by: Balaji B
RegisterNumber: 212225040040
'''
import os
os.environ['OPENBLAS_NUM_THREADS']="1"
import numpy as np
import sys
n=int(input())
a=np.zeros((n,n+1))
x=np.zeros(n)
for i in range(n):
    for j in range(n+1):
        a[i][j]=float(input())
for i in range(n):
    if a[i][i]==0:
        sys.exit('Divide by Zero Detected!')
    for j in range(i+1,n):
        ratio=a[j][i]/a[i][i]
        for k in range(n+1):
            a[j][k]=a[j][k]-ratio*a[i][k]
x[n-1]=a[n-1][n]/a[n-1][n-1]
for i in range(n-2,-1,-1):
    x[i]=a[i][n]
    for j in range(i+1,n):
        x[i]=x[i]-a[i][j]*x[j]
    x[i]=x[i]/a[i][i]
for i in range(n):
    print('X%d = %0.2f'%(i,x[i]),end=' ')
*/
```

## Output:

Use Gaussian elimination without partial pivoting to solve a matrix.

Hint: First value is the number of unknowns, remaining values are the elements of the matrix.

**For example:**

| Input | Result |
|---|---|
| 3<br>1<br>2<br>4<br>18<br>2<br>12<br>-2<br>9<br>5<br>26<br>5<br>14 | X0 = 53.35 X1 = -8.88 X2 = -4.40 |

**Answer:** (penalty regime: 0 %)

Reset answer

```
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import numpy as np
import sys
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x=np.zeros(n)
for i in range(n):
    for j in range(n+1):
        a[i][j]=float(input())
```

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